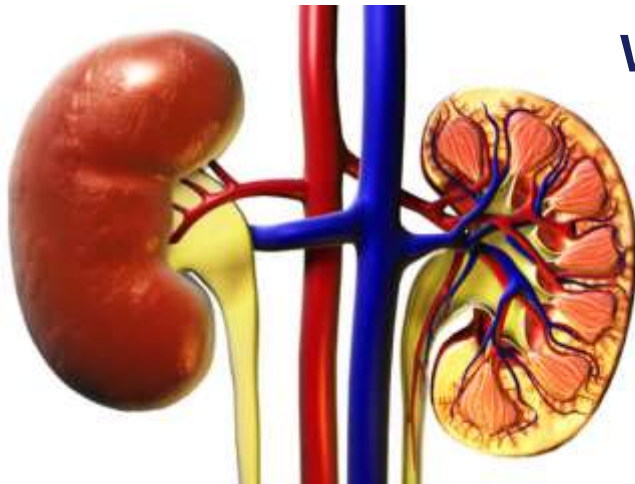


Diuretics



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Nephron: Urine formation

- Nephron:**

- **Glomerulus:** Glomerular capillaries + Bowman's capsule
- **Renal tubule:** Proximal tubule + Loop of Henle + Distal tubule + Collecting duct

- Urine formation**

Blood in afferent arterioles



Glomerulus (renal corpuscle)

Filtration



Glomerular filtrate



Renal tubules

Reabsorption & Excretion



Ureter

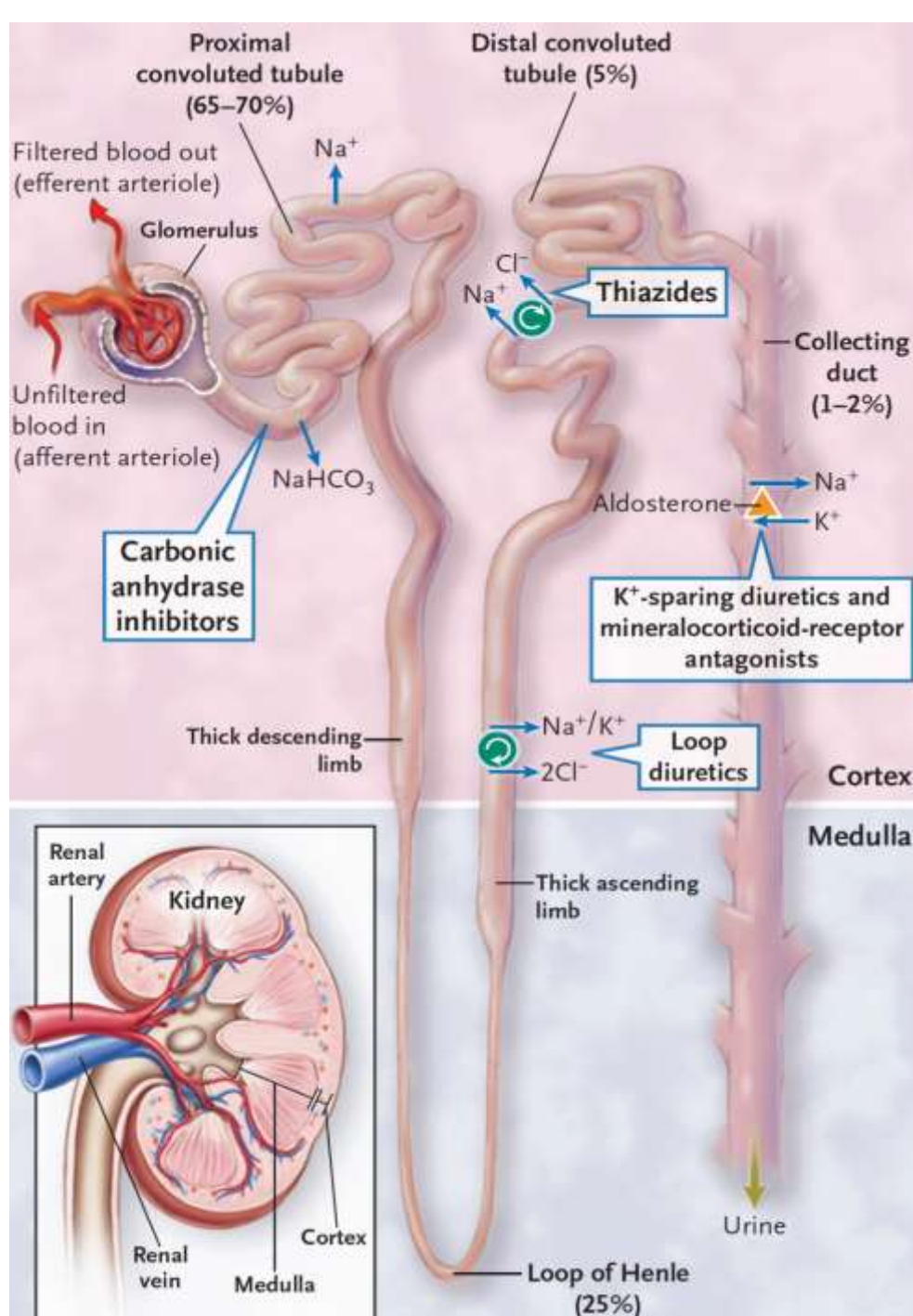


Bladder



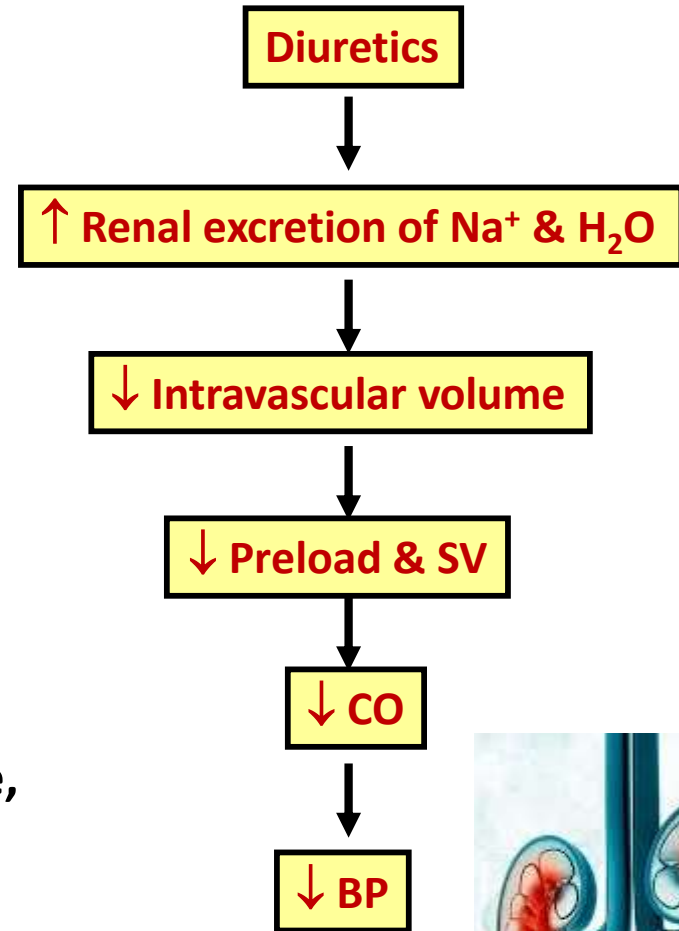
Urine

2



Diuretics

- **Diuretic:** is an agent that promotes **diuresis**, the increased production of urine (increases urine volume)
 - ↑ Excretion of excessive water and electrolytes (mainly Na^+)
 - ↓ Reabsorption of excessive water and electrolytes at renal tubule
- **Natriuretic:** ↑ Renal sodium excretion
- **Aquaretic:** ↑ Excretion of solute-free water
- Diuretics are used to treat heart failure, kidney disease, liver cirrhosis, hypertension



Classification of diuretics

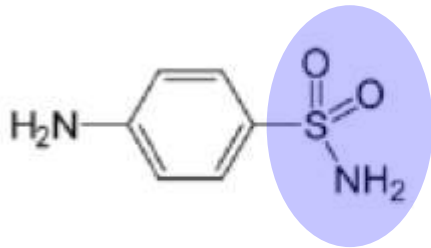
Diuretic	Target of action	Mechanism
Carbonic anhydrase inhibitor	Proximal convoluted tubule (PCT)	Inhibit carbonic anhydrase
Thiazides diuretic & thiazide-like diuretic	Distal convoluted tubule (DCT)	Inhibit $\text{Na}^+\text{-Cl}^-$ symporter
Loop diuretic	Thick ascending limb of loop of Henle	Inhibit $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter
K^+-sparing agent	Late distal tubule, collecting duct	• Na^+ channel inhibitor • Aldosterone receptor antagonist
Osmotic diuretic	Proximal tubule, loop of Henle, collecting tubule	Osmotic effect

Changes in urinary electrolyte patterns and body pH

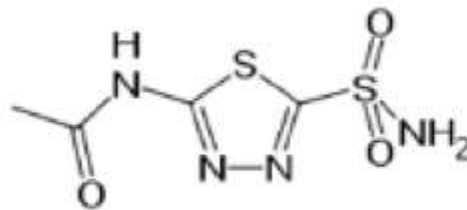
Diuretic	Urinary electrolyte			Blood pH
	NaCl	NaHCO ₃	K ⁺	
Carbonic anhydrase inhibitor	+	+++	+	↓
Thiazides & thiazide-like diuretic	++	+	+	↑
Loop diuretic	++++	0	+	↑
K ⁺ -sparing agent	+	+	-	↓
Osmotic diuretic	+	+	+	N/A

Carbonic anhydrase inhibitor (CAI)

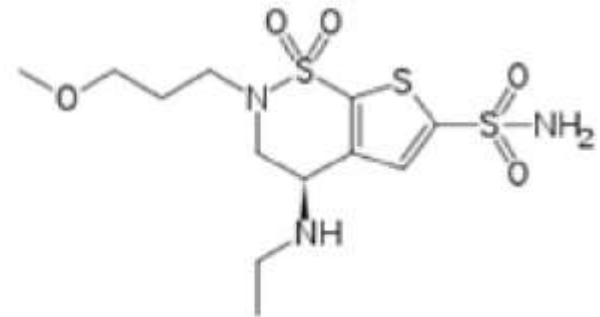
CAI: Structure



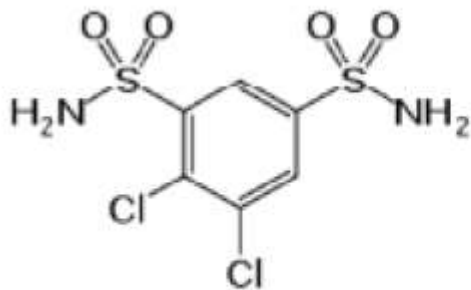
Sulfanilamide



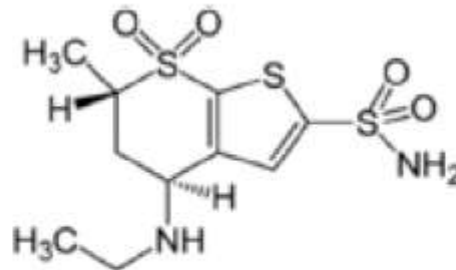
Acetazolamide



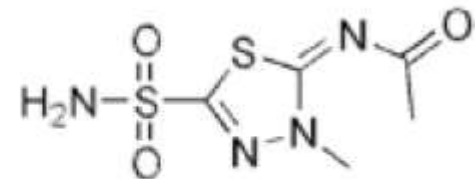
Brinzolamide



Dichlorophenamide



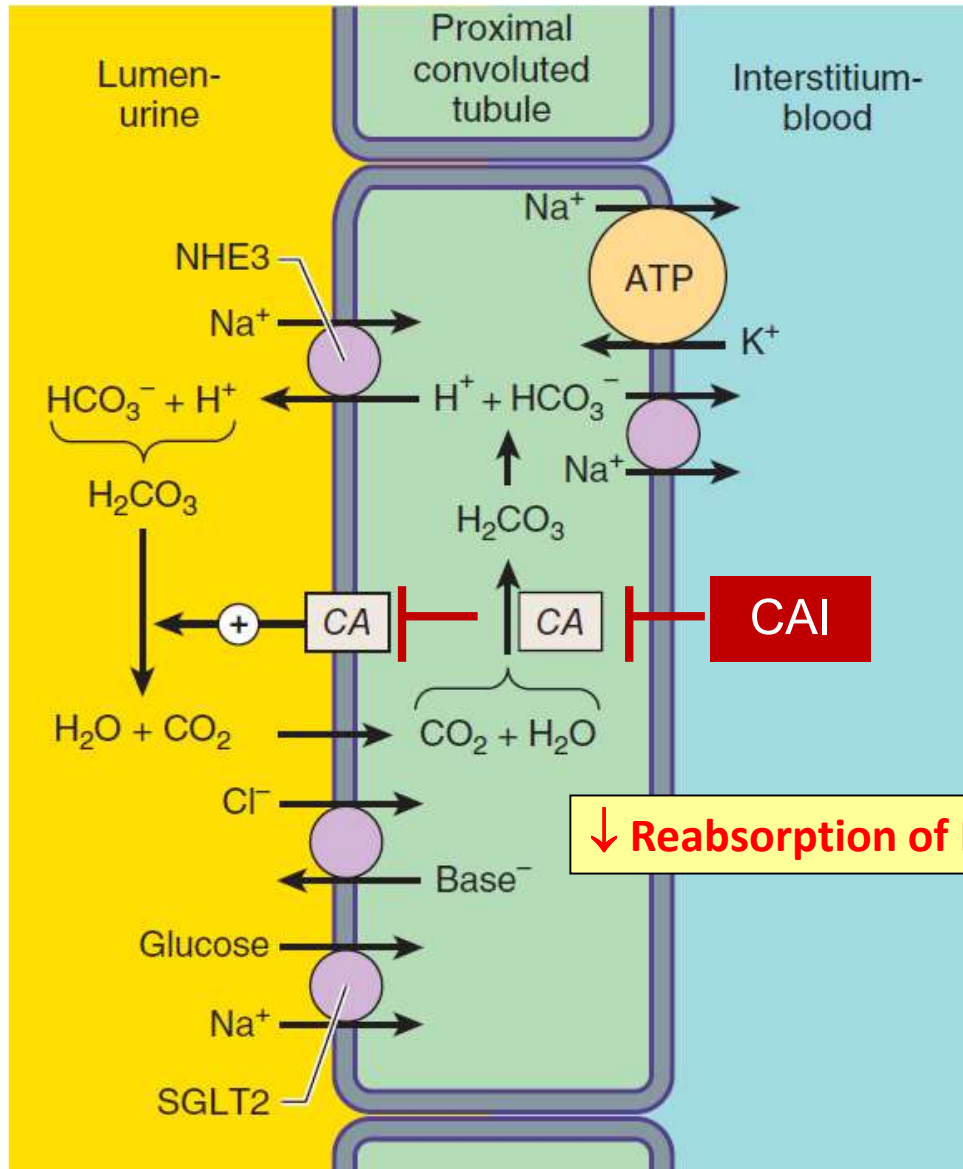
Dorzolamide



Methazolamide

CAI: Sulfonamide-derivative structure

CAI: Mechanism of action



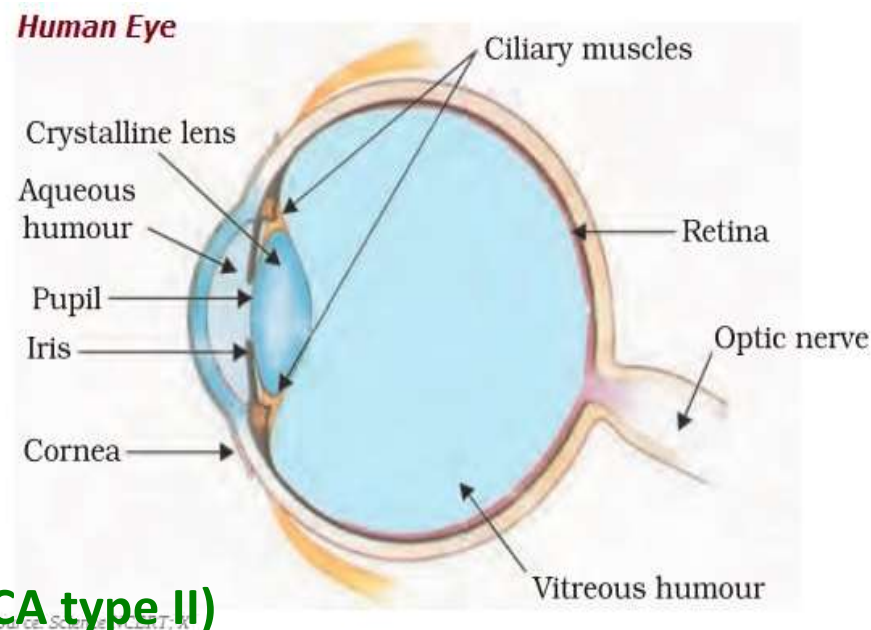
Proximal convoluted tubule

- Mechanism:** Inhibit carbonic anhydrase (CA), prevents dehydration of H₂CO₃ and hydration of CO₂ in proximal convoluted tubule
- $$\text{CO}_2 + \text{H}_2\text{O} \xrightleftharpoons{\text{CA}} \text{H}_2\text{CO}_3 \rightarrow \text{HCO}_3^- + \text{H}^+$$
- CA type IV → membrane
 - CA type II → cytosol

CAI: Pharmacological effects

- Effects on electrolytes & renal hemodynamics:

- $\uparrow \text{Na}^+$, HCO_3^- and H_2O excretion
- $\uparrow \text{K}^+$ excretion (at collecting duct)
- $\downarrow$ Ammonia excretion
- Metabolic acidosis
- Urine: $\uparrow$ pH



- Effects on eye: (Ciliary epithelial cell - CA type II)

- $\downarrow \text{HCO}_3^-$ synthesis $\rightarrow$ $\downarrow$ Aqueous humor $\rightarrow$ $\downarrow$ Intraocular pressure (IOP)

Use for Glaucoma

- Effects on other tissues:

- CA is found at other organs: CNS, pancreas, red blood cells
- CNS: ADR: Paresthesia, somnolence

CAI: Pharmacokinetics

CARBONIC ANHYDRASE INHIBITORS (CAI) COMPARISON CHART

	Trade name	Dosage form	Onset	Peak	Duration	Half-life	Relative Potency
CAI Topical							
Brinzolamide	Azopt	Ophth Susp 1%	< 2 hr	2 hr	>12 hr	111 days	
Dorzolamide	Trusopt	Ophth Soln 2%	< 2 hr	2-4 hr	6-8 hr	147 days	
CAI Oral							
Acetazolamide	Diamox	Tab 125, 250 mg	1-2 hr	2-4 hr	4-12 hr	6-9 hr	1
	Diamox Sequel	SR Cap 500 mg	2-4 hr	8 hr	12-24 hr		
Acetazolamide Na		Inj 500 mg vial	2 min	15 min	4-5 hr		
Dichlorphenamide	Daranide	Tab 50 mg	30 min	2-4 hr	6-12 hr	2 hr	30
Methazolamide	Naptazane	Tab 25, 50 mg	1-2 hr	4-6 hr	12-24 hr	14-15 hr	>1 & <10

CAI: Adverse effects

- Allergic reactions: Do not use in patients with sulfonamide hypersensitivity
- Metabolic acidosis
 - Chronic use of CAI -> Chronic reduction of body HCO_3^- stores
- Urinary alkalization:
 - Precipitate calcium phosphate -> calculus formation (renal calculi)
- Renal potassium wasting:
 - Increased NaHCO_3^- is reabsorbed in collecting tubule -> $\uparrow \text{K}^+$ excretion
 - Prevention: Co-administer with potassium chloride or a K^+ -sparing diuretic
- Hyperammonemia in cirrhotics
- Others: Paresthesia, somnolence, depression, fatigue

CAI: Therapeutic Uses

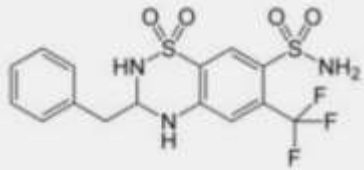
- **Glaucoma:** Acetazolamide, Brinzolamide, Dorzolamide
- **Metabolic alkalosis:** Acetazolamide
- **Acute mountain (altitude) sickness :** Acetazolamide (↓ CSF formation in brain)

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Acetazolamide (Diamox®)	Glaucoma: 250-1000 mg/day (qid) Prevention of mountain sickness Metabolic alkalosis	Tab 250 mg	ก
Brinzolamide (Azopt®)	Glaucoma	Eye drop 1%	ค
Brinzolamide + Timolol (Azarga®)	Glaucoma	Eye drop	Non-ED
Dorzolamide (Trusopt®)	Glaucoma	Eye drop 2%	-
Dorzolamide + Timolol (Cosopt®)	Glaucoma	Eye drop	Non-ED

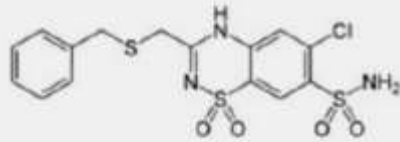
**Thiazide diuretic &
thiazide-like diuretic
("Thiazide")**

Thiazide: Structure

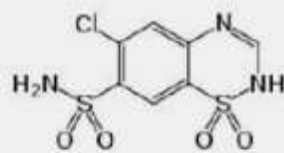
Thiazide diuretics



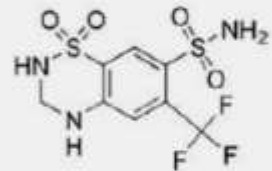
Bendroflumethiazide



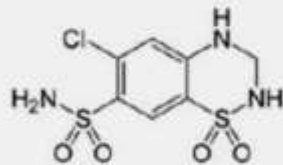
Benzthiazide



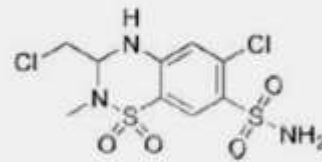
Chlorothiazide



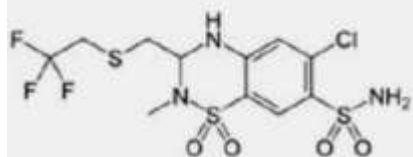
Hydroflumethiazide



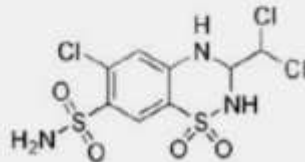
Hydrochlorothiazide



Methyclothiazide

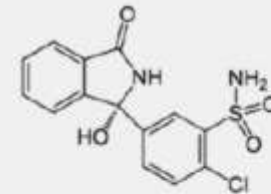


Polythiazide

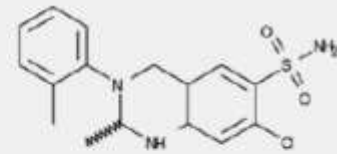


Trichlormethiazide

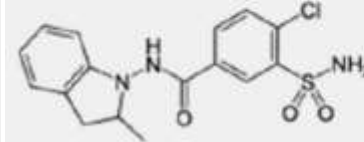
Thiazide-like diuretics



Chlorthalidone



Metolazone

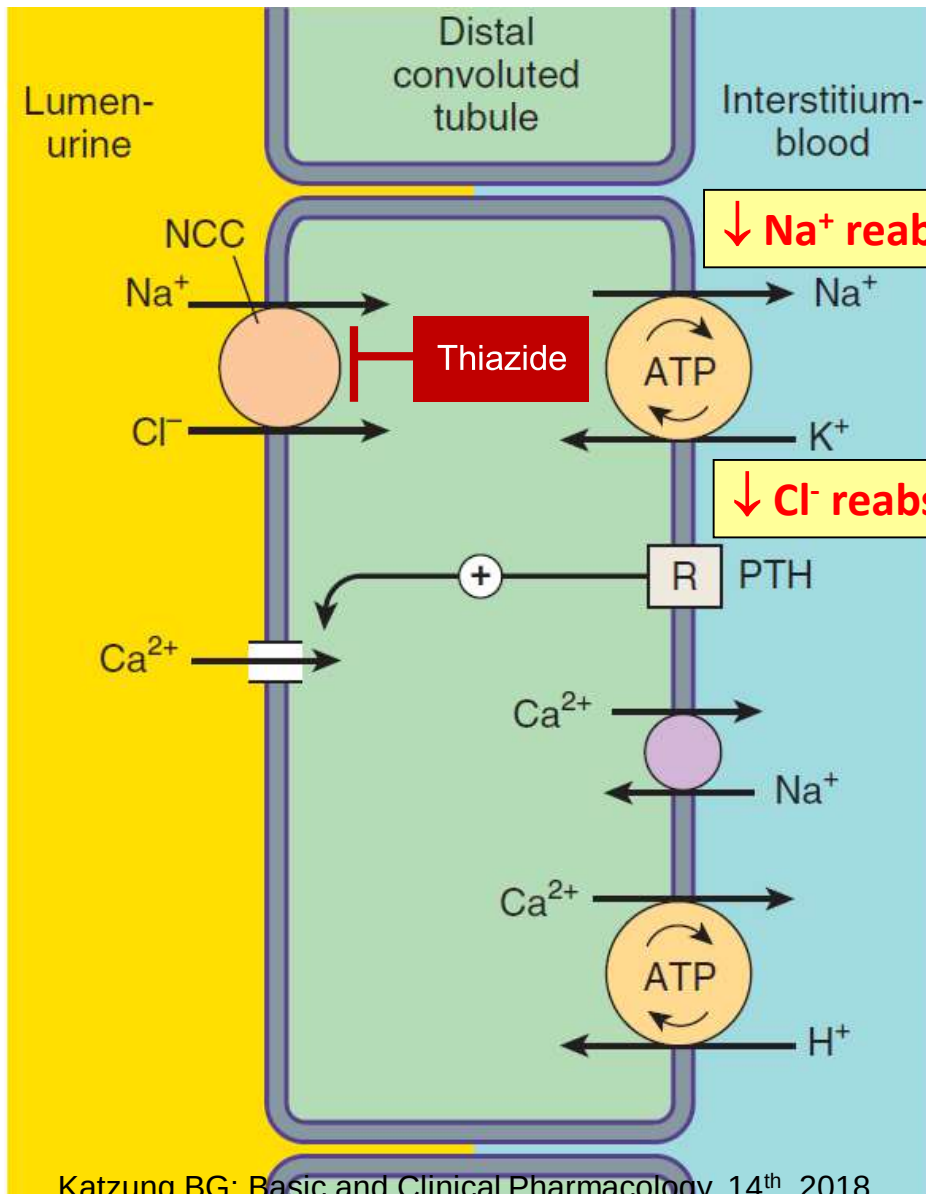


Indapamide

Thiazide: Sulfonamide-derivative structure

Both thiazide and thiazide-like is similar in pharmacological action and adverse effect.
But thiazide-like was developed to have a longer duration of action

Thiazide: Mechanism of action



Distal convoluted tubule

- **Mechanism: Inhibit Na⁺-Cl⁻ symporter** in the luminal membrane of distal convoluted tubule

Pharmacological effects

- **Effects on electrolytes & renal hemodynamics:**
 - **↑ Na⁺ and Cl⁻ excretion**
 - **↑ K⁺ excretion** (at collecting duct)
 - **↓ Uric acid excretion**

Thiazide: Pharmacokinetics

THIAZIDE AND THIAZIDE-LIKE DIURETICS COMPARISON CHART

	Trade name	Dosage form	Onset	Peak	Duration	Half-life	Relative Potency
Bendroflumethiazide	Naturetin	Tab 5, 10 mg	PO 2 hr	4 hr	6-12 hr	3-4 hr	10
Benzthiazide	Exna	Tab 50 mg	PO 2 hr	4-6 hr	12-18 hr	ID	1
Chlorothiazide	Diuril	Tab 250, 500 mg	PO 2 hr	4 hr	6-12 hr	0.5-2 hr	0.1
		Susp 50 mg/ml	IV 15 min	0.5 hr	2 hr		
		Inj 500 mg					
Chlorthalidone	Hygroton	Tab 25, 50, 100 mg	PO 2-3 hr	2 hr	24-72 hr	40 hr	1
Hydrochlorothiazide		Cap 12.5 mg	PO 2 hr	4-6 hr	6-12 hr	6-15 hr	1
		Tab 25, 50, 100 mg					
		Soln 10 mg/ml					
Hydroflumethiazide	Diurcardin	Tab 50 mg	PO 2 hr	3-4 hr	12-24 hr	17 hr	1
Indapamide	Lozol	Tab 1.25, 2.5 mg	PO 1-2 hr	2 hr	up to 36 h	14 hr	20
Methyclothiazide	Aquatensen	Tab 2.5, 5 mg	PO 2 hr	6 hr	24 hr	ID	10

Thiazide: Adverse effects

- **Electrolyte abnormalities:**
 - **Hypokalemia****, hyponatremia, hypochloremia, hypomagnesemia
 - **Prevention:** K⁺-supplement or K⁺-sparing diuretic
- **Hyperuricemia** (Caution: **gout**)
- **Impaired glucose tolerance** (Caution: **DM**)
- **↑ cholesterol & TG level** (Caution: **hyperlipidemia**)
- **Ventricular rhythm disturbance:** Due to hypokalemia
- Rarely: CNS (vertigo, headache), GI (nausea, vomiting), etc

Thiazide: Cautions

- **Do not use in patients with sulfonamide hypersensitivity**
- **Drug Interactions:**
 - Thiazides: ↓ effects of **uricosuric agent, sulfonylurea** and **insulin**
 - **NSAIDs:** ↓ renal blood flow -> ↓ Diuretic effect
 - **Cardiac glycoside (digoxin):** ↑ risk of digoxin toxicity (from hypokalemia) **(Due to thiazide)****

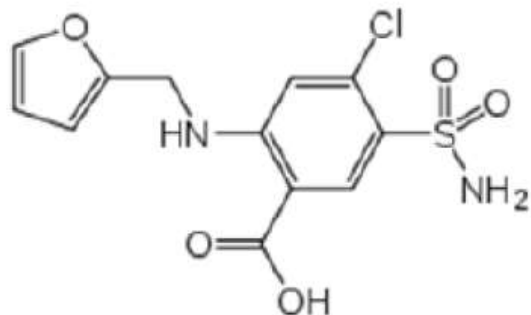
Thiazide: Therapeutic uses

- **Hypertension:** HCTZ, indapamide
 - For mild hypertension, As monotherapy or combination therapy (with ACEIs, ARBs, β -blockers)
- **Edema:** HCTZ, indapamide
 - For edema in many diseases: Heart failure, cirrhosis, renal insufficiency

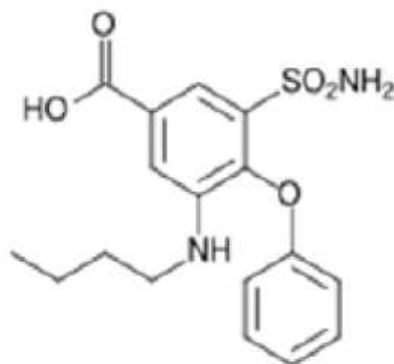
Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Hydrochlorothiazide (HCTZ)	Diuretic (Hypertension, Edema)	Tab 25, 50 mg Syrup 10 mg/mL	ก
Chlorthalidone +Atenolol (Tenoretic®)	Diuretic (Hypertension, Edema)	Tab 12.5/50 mg	-
Indapamide (Natrlix®)	Diuretic (Hypertension, Edema)	Tab 2.5 mg	Non-ED

**Loop diuretic
(High ceiling)**

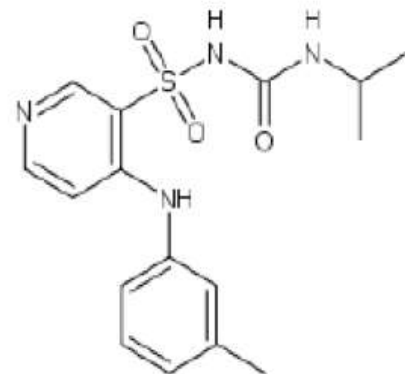
Loop diuretic: Structure



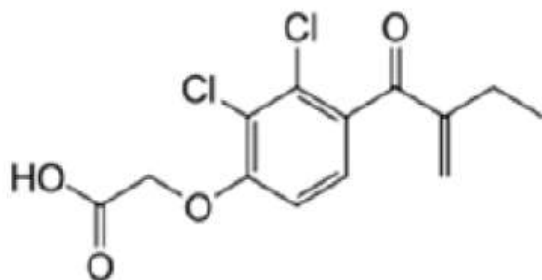
Furosemide



Bumetanide



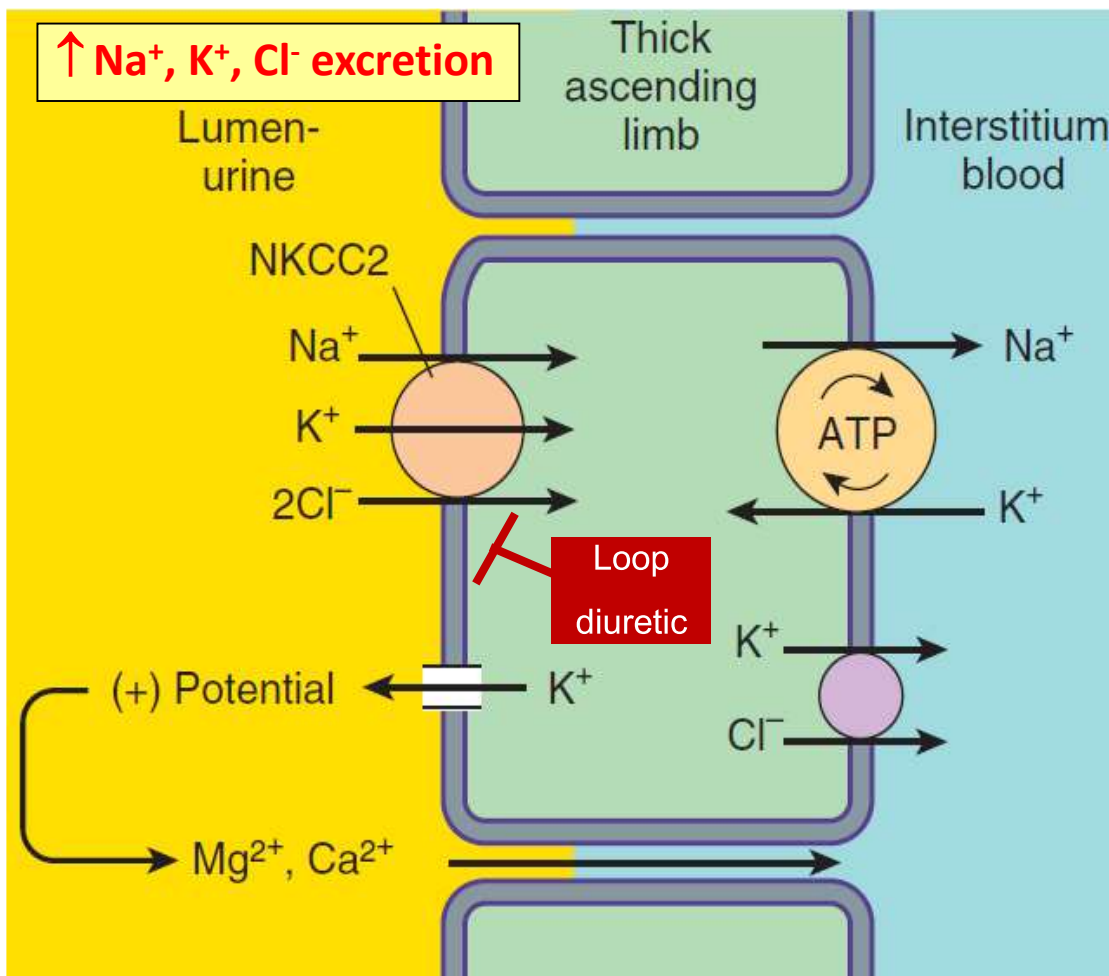
Torasemide



Ethacrynic Acid

Furosemide, bumetanide and torasemide are sulfonamide-derivative structure, while ethacrynic acid is a phenoxyacetic acid derivative

Loop diuretic: Mechanism of action



Furosemide, bumetanide, torasemide: act on Cl^- -binding site of symporter

Ethacrynic acid: form complex with cysteine residue to be active form

Loop of Henle

- Mechanism:** Block $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter in ascending limb of Henle's

Pharmacological effects

- Effects on electrolytes & renal hemodynamics:**
 - $\uparrow \text{Na}^+, \text{K}^+, \text{Cl}^-$ excretion
 - $\downarrow$ Uric acid excretion
- Hemodynamic action**
 - Venodilator**, through release of **nitric oxide (NO)**

Loop diuretic: Pharmacokinetics

LOOP DIURETICS COMPARISON CHART

	Trade name	Dosage form	Onset	Peak	Duration	Half-life	Relative Potency
Furosemide	Lasix	Tab 20, 40, 80 mg	PO 20-60 min	1-2 hr	6-8 hr	1 hr	1
		Oral solu 8, 10 mg/ml	IV in 5 min	30 min	2 hr		
		Inj 10 mg/ml	IM Unknown	30 min	2 hr		
Ethacrynic Acid	Edecrin	Tab 25, 50 mg	PO 30 min	2 hr	6-8 hr	1 hr	0.7
		Inj 50 mg	IV 5 min	15-30 min	2 hr		
Bumetanide	Bumex	Tab 0.5, 1, 2 mg	PO 30-60 min	1-2 hr	4-6 hr	1-1.5 hr	40
		Inj 0.25mg/ml	IV in 5 min	15-30 min	0.5-1 hr		
			IM 40 min	unknown	5-6 hr		
Torsemide	Demadex	Tab 5, 10, 20, 100 mg	PO 1 hr	1-2 hr	6-8 hr	3-4 hr	3
		Inj 10 mg/ml	IV 10 min	1 hr	6-8 hr		

Loop diuretic: Adverse effects

- **Electrolyte abnormalities**
 - **Hypokalemia****, hyponatremia, hypochloremia, hypomagnesemia
- **Hypokalemic metabolic alkalosis**
- **Hyperuricemia** (Caution: **gout**)
- **Impaired glucose tolerance** (Caution: **DM**)
- **↑ Cholesterol & TG levels** (Caution: **hyperlipidemia**)
- **Ototoxicity****: tinnitus, hearing impairment, vertigo, deafness
 - Rapid IV injection of large dose
- **Dehydration, hypotension**

Loop diuretic: Cautions

- **Drug interactions:**
 - Loop diuretics: ↓ effects of **uricosuric agent, sulfonylurea and insulin**
 - **NSAIDs**: ↓ Diuretic effect of loop diuretics
 - **Aminoglycosides or Cisplatin**:
↑ Ototoxicity

Loop diuretic: Therapeutic uses

- **Hypertension** (not first-choice)
- **Edema**: Heart failure, cirrhosis, nephritic syndrome
- **Acute pulmonary edema, peripheral edema**
- **Chronic renal failure**
- **Hyperkalemia**

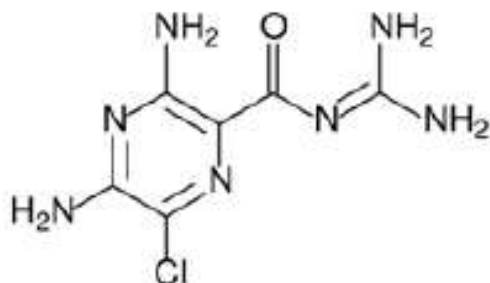
Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Furosemide (Lasix®)	Diuretic	Tab 40, 500 mg Injection 20mg/2mL, 250mg/25mL Syrup 2mg/mL	ก
Torsemide (Unat®)	Diuretic	Tab 10 mg	Non-ED

K⁺-sparing diuretic

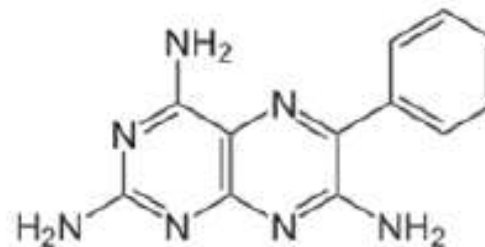
- Na⁺ channel inhibitor (NCC)
- Aldosterone receptor antagonist (ARA)
(or Mineralocorticoid receptor antagonist; MRA)

K⁺-sparing diuretic: Structure

1. Na⁺ channel inhibitor (NCC): Amiloride, Triamterene

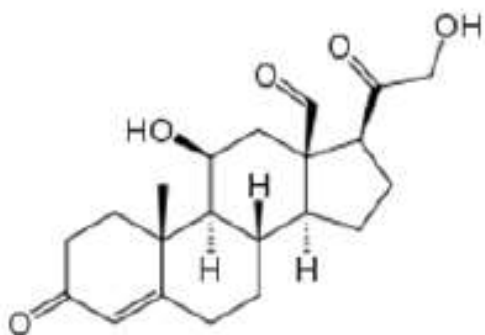


Amiloride

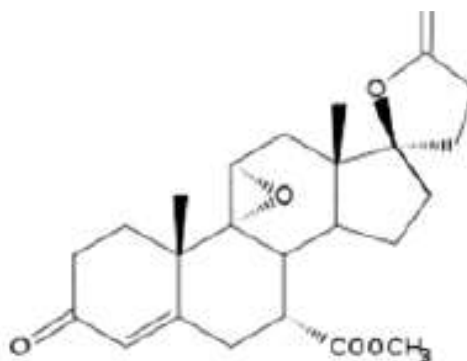


Triamterene

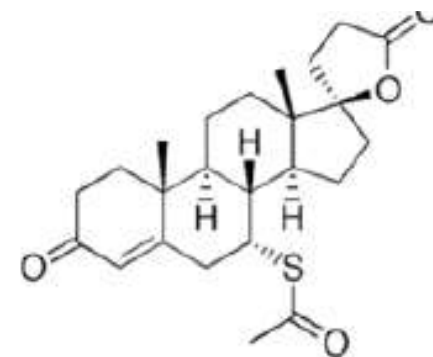
2. Aldosterone antagonist (ARA): Spironolactone, Eplerenone



Aldosterone

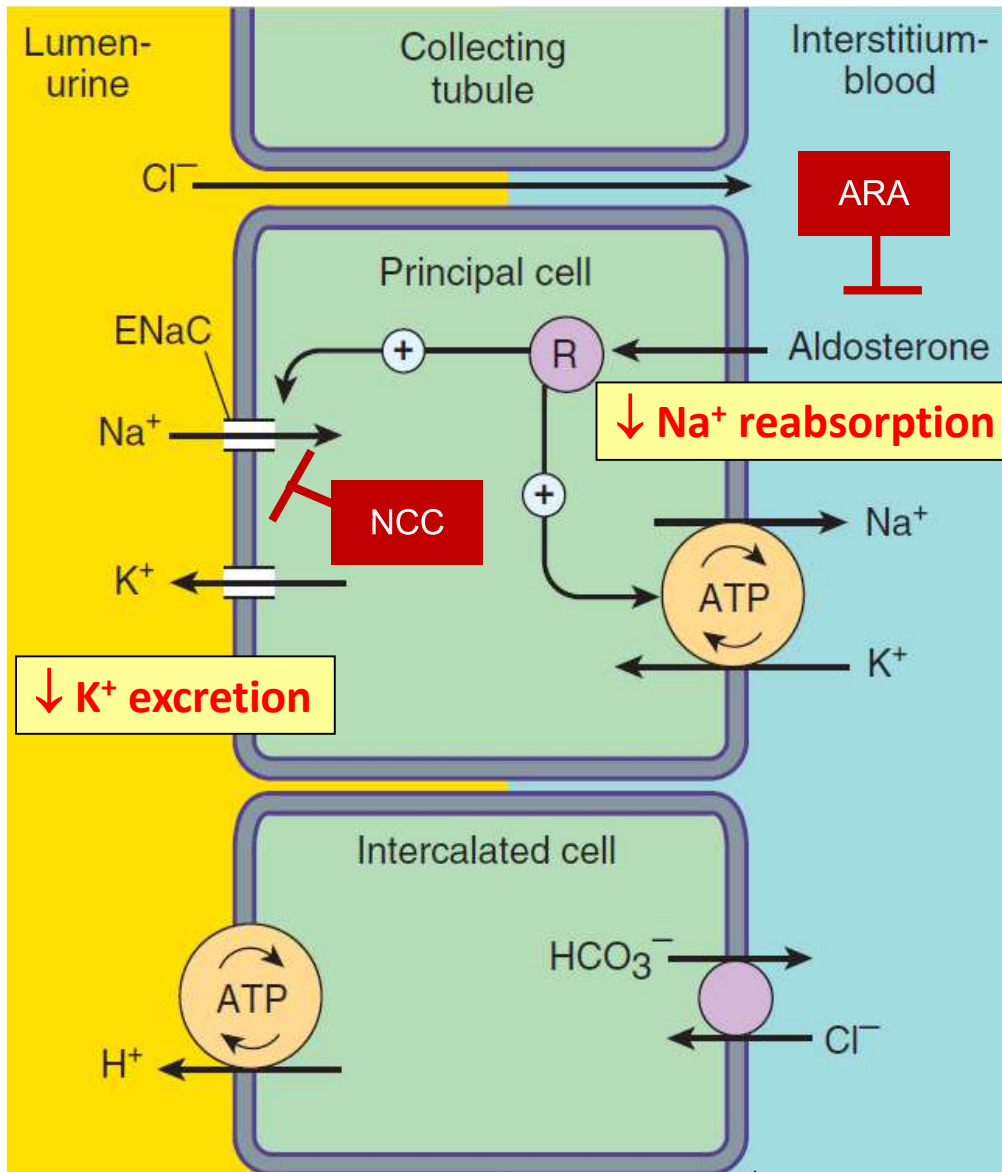


Eplerenone



Spironolactone

K⁺-sparing diuretic: Mechanism of action



Cortical collecting tubule

- **NCC: Block Na⁺ channel** at distal tubule & collecting duct → ↓ Na⁺ reabsorption & ↓ K⁺ excretion
- **ARA: Antagonize aldosterone action** → ↓ Na⁺ reabsorption & ↓ K⁺ excretion (Aldosterone ≡ receptor → ↑ activity of Na⁺ channel, Na⁺-K⁺ ATPase)

Pharmacological effects

- **Effect on electrolyte & renal hemodynamics**
 - ↑ Na⁺ and Cl⁻ excretion
 - ↓ K⁺ excretion → **Hyperkalemia**

K⁺-sparing diuretic: Pharmacokinetics

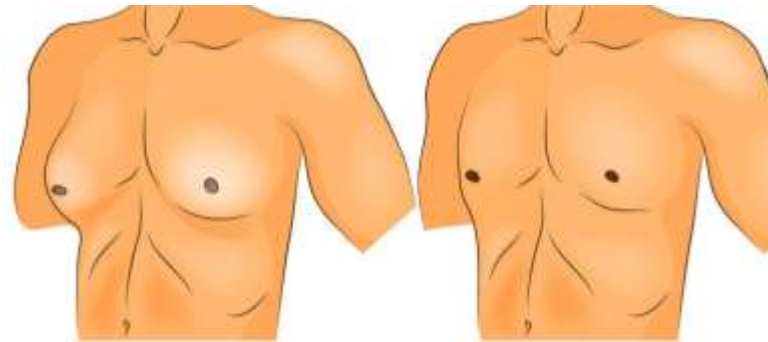
K⁺-SPARING DIURETICS COMPARISON CHART

	Trade name	Dosage form	Onset	Peak	Duration	Half-life	Relative Potency
Amiloride	Midamor	Tab 5 mg	PO 2 hr	6-10 hr	24 hr	6-9 hr	1
	Moduretic	Tab 5 mg with HCTZ 50 mg					
Triamterene	Dyrenium	Cap 50, 100 mg	PO 2-4 hr	6-8 hr	12-16 hr	3 hr	0.1
	Dyazide	Tab 50 mg with HCTZ 25 mg					
Spirolonactone	Aldactone	Tab 25, 50, 100 mg	PO 1-2 day	2-3 day	2-3 day	20 hr	ID

ID = insufficient data

K⁺-sparing diuretic: Adverse effects

- **Electrolyte abnormalities**
 - **Hyperkalemia**** (especially in renal impairment)
 - Dehydration, hyponatremia (occasionally)
 - Drug interaction: ACEIs, K⁺ supplement: ↑ risk of hyperkalemia



- **Metabolic acidosis**
- **GI disturbance** (nausea, vomiting, diarrhea)
- **Spironolactone**: is steroid-structure -> bind **other steroid receptors**
(androgen receptors and progestogen receptors)
 - Male: **Gynecomastia, impotence** (does not occur with Eplerenone)
 - Female: **Menstrual irregularation, hair growth**

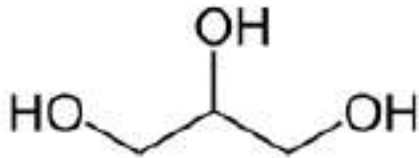
K⁺-sparing diuretic: Therapeutic uses

- **Hypertension:** Amiloride, Spironolactone
- **Edema:** Amiloride, Spironolactone
- **Primary hyperaldosteronism (Conn's syndrome):** Spironolactone
- **Heart failure:** Spironolactone, Eplerenone

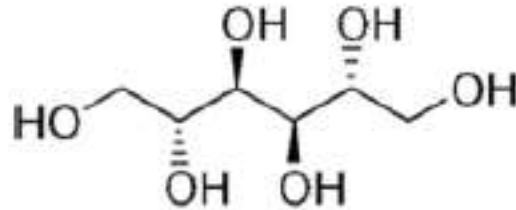
Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Spironolactone (Aldactone®)	Diuretic (Hypertension, Edema)	Tab 25, 100 mg Syrup 2mg/mL	ก
Amiloride +HCTZ (Moduretic®)	Diuretic (Hypertension, Edema)	Tab 5/50 mg	ข

Osmotic diuretic

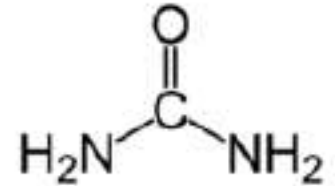
Osmotic diuretic: Structure



Glycerin



Mannitol



Urea

- **Properties:**

- Inert pharmacologically substance
- **Freely filtered at glomerulus (almost completed)**
- **No reabsorption at renal tubule**

Osmotic diuretic: Mechanism of action

Mechanism:

- Osmotic agent **is filtered by glomerulus (but not reabsorbed)**
- **Retain water in renal tubule** -> $\uparrow$ urine volume **"Aquaretic"**

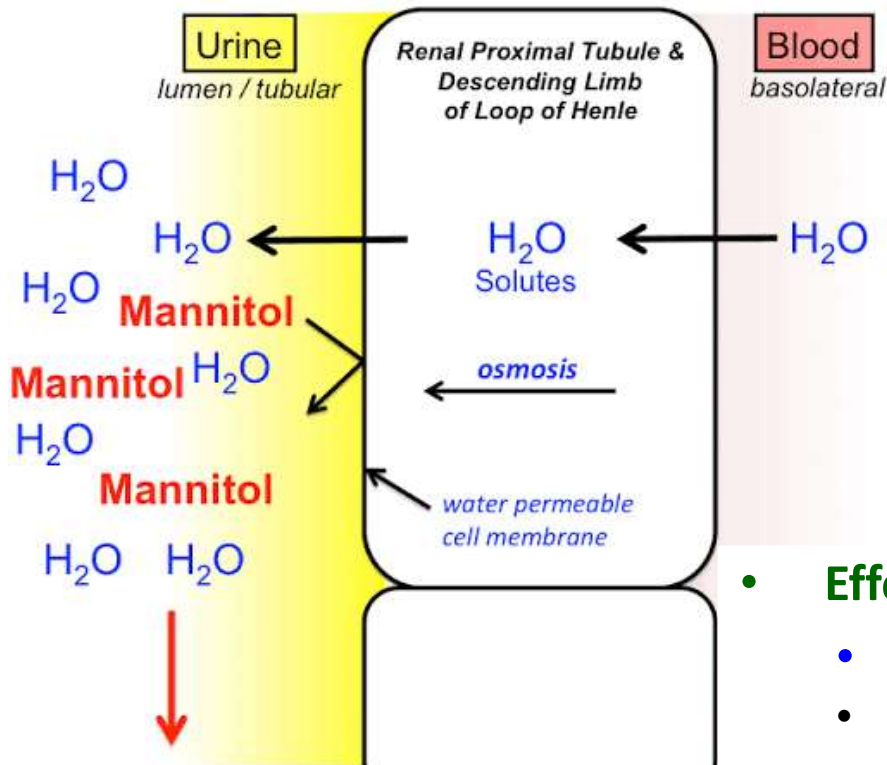
Pharmacological effects

- **Effects on electrolytes & renal hemodynamics**
 - **Major:** $\uparrow$ **Urine volume & flow rate**
 - **Minor:** $\uparrow$ Excretion of electrolytes in urine: Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- , phosphate

- $\uparrow$ **Serum osmolality:**

Shifts fluid out of eye $\rightarrow$ $\downarrow$ **Intraocular pressure**

Shifts fluid from brain $\rightarrow$ $\downarrow$ **Cerebral edema**



Osmotic diuretic: Pharmacokinetics

OSMOTIC DIURETICS COMPARISON CHART

Trade name		Dosage form	Onset	Peak	Duration	Half-life	Ocular penetration
Glycerin			PO 10-30 min	1-1.5 hr	4-5 hr	30-45 min	Poor
Isosorbide			PO 10-30 min	1-1.5 hr	5-6 hr	5-10 hr	Good
Mannitol	Osmitrol	Inj 5, 10, 15, 20, 25%	IV 30-60 min	1 hr	6-8 hr	1-2 hr	Very poor
Urea			IV 30-45 min	1 hr	5-6 hr	ID	Good

ID = insufficient data

Osmotic diuretic: Adverse effects

- **Dehydration: headache, nausea, vomiting**
 - Osmotic diuretic: $\uparrow$ ECF osmolality
-> Remove water from intracellular compartment -> $\uparrow$ ECF volume

Osmotic diuretic: Cautions

- **Precaution:**
 - **Heart failure, pulmonary congestion**
- **Contraindication:**
 - **Anuria, severe renal diseases**
 - **Active intracranial bleeding**
 - **Severe pulmonary congestion/edema**
 - **Severe HF**

Osmotic diuretic: Therapeutic uses

- **Maintain RBF (Prevent oliguria or acute renal failure): Mannitol**
- **Reduction of intraocular pressure: Mannitol**
- **Reduction of intracranial pressure (Reduce cerebral edema): Mannitol**

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Mannitol	Diuretic Reduction of intracranial & intraocular pressure	IV	ก

Clinical applications of diuretics



Clinical applications

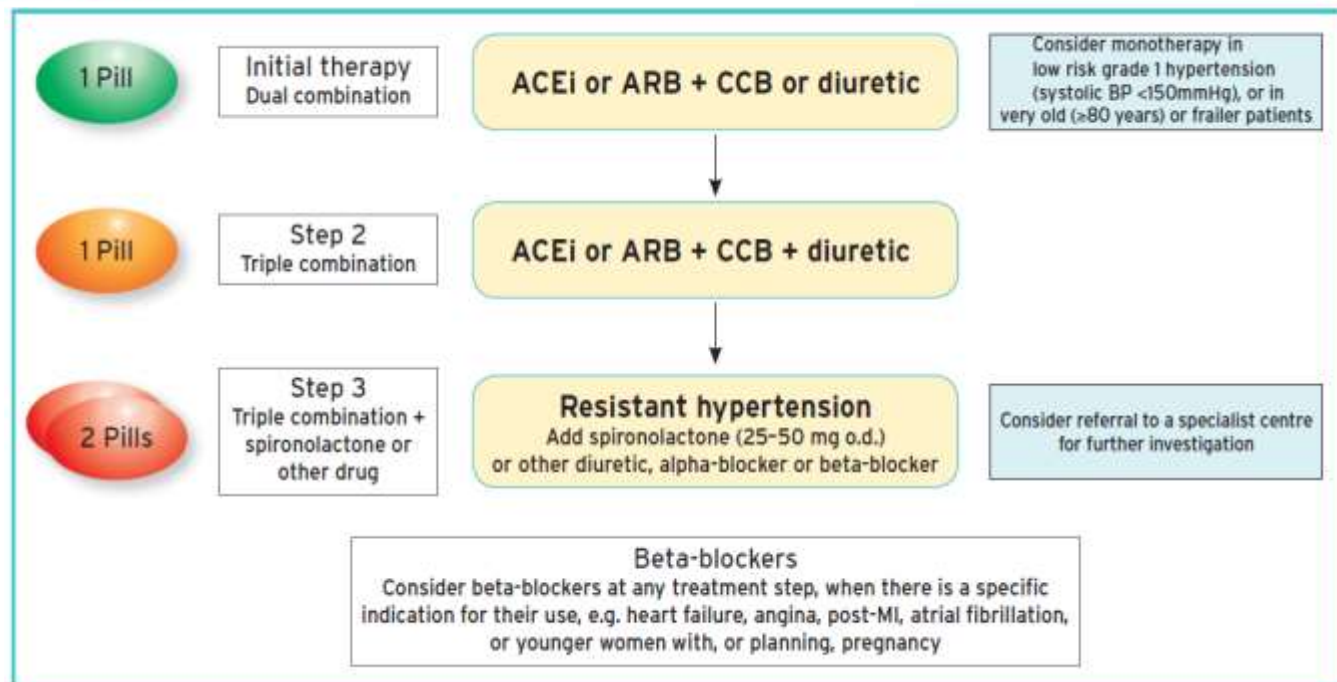
- **Edematous state**
 - Heart failure
 - Kidney disease, renal failure
 - Hepatic cirrhosis
- **Non-edematous state**
 - Hypertension
 - Nephrolithiasis
 - Renal & cardiac protection



Example

2018 ESC/ESH Guidelines for the management of arterial hypertension

For hypertension



For hypertension with heart failure (HFrEF)

